

**Quiz 5**

Oct 1st, 2024

Time: 15 minutes

Name: \_\_\_\_\_ Roll no: \_\_\_\_\_ Group: \_\_\_\_\_

Instructions: Notes, books, computers, cell phones and other electronic devices are not allowed.  
Max marks = 5.

**Problem 1.** 1. Determine whether the following vectors form a linearly independent set.  $v_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \\ 1 \end{bmatrix}$

$$, v_2 = \begin{bmatrix} -2 \\ 1 \\ -6 \\ -4 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ -6 \\ 3 \\ -9 \end{bmatrix}.$$

2. Does the vector  $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$  belong to  $\text{Span} \{v_1, v_2, v_3\}$ ? Show reasons.